

# Project 2 - AWS S3 Bucket Management Report

## Introduction

Cloud computing provides scalable and on-demand storage solutions. Amazon S3 (Simple Storage Service) is a widely used object storage service offered by AWS that allows users to store and retrieve data over the internet. It is designed for high durability, availability, and security.

This project focuses on understanding the fundamental operations of Amazon S3, including bucket creation, object management, and resource deletion.

## Objective

The main objective of this project is to:

1. Create an S3 bucket
2. Upload and manage files (objects)
3. Retrieve and download stored objects
4. Delete objects and the bucket
5. Understand the lifecycle of cloud storage resources

## Overview of Amazon S3

Amazon S3 is an object storage service where data is stored in the form of objects within containers called buckets.

### Key Concepts:

1. **Bucket:** A container for storing objects; must have a globally unique name
2. **Object:** A file along with its metadata stored in a bucket
3. **Region:** Physical location where the bucket is hosted
4. **Key:** Unique identifier for an object within a bucket

## Methodology

### Bucket Creation

A new S3 bucket was created using the AWS Management Console. A unique bucket name was provided, and a region was selected. Default security settings were applied, including blocking all public access.

### Uploading Objects

Files were uploaded into the created bucket. These files are treated as objects in S3, each having a unique key and associated metadata such as size and last modified date.

## Accessing and Downloading Objects

Uploaded objects were accessed through the S3 console. The objects were successfully downloaded to verify retrieval functionality.

## Deleting Objects

All uploaded objects were deleted from the bucket. This step is necessary because AWS does not allow deletion of a non-empty bucket.

## Deleting the Bucket

After ensuring that the bucket was empty, the bucket was permanently deleted from the S3 service.

## Results

1. Successfully created an S3 bucket
2. Uploaded and managed multiple objects
3. Retrieved and downloaded stored files
4. Performed deletion of objects and bucket
5. Demonstrated full lifecycle of S3 storage operations

## Key Learnings

1. Amazon S3 uses object-based storage instead of file systems
2. Bucket names must be globally unique across AWS
3. Data in S3 is highly durable and scalable
4. Security settings such as blocking public access are enabled by default
5. Buckets must be empty before deletion

## Advantages of Amazon S3

1. Highly scalable storage
2. 99.999999999% (11 9's) durability
3. Secure with IAM and bucket policies
4. Cost-effective pay-as-you-go model
5. Easy integration with other AWS services

## Limitations

1. Requires internet access
2. Costs can increase with high data transfer
3. Misconfigured permissions can lead to security risks

## Conclusion

This project provided practical exposure to Amazon S3 and its core functionalities. It helped in understanding how cloud storage works, how data is managed in object storage systems, and the importance of security and proper resource management in cloud environments.